

LESSON 134 (1.5 GROUND)**Lesson Objective**

The purpose of this lesson is to introduce night VFR cross-country planning, procedures and regulations applicable to a night VFR cross-country flight.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Review night operations as this topic will be quizzed on next lesson.
 - Review cross-country flight planning procedures.
 - Review emergencies.
 - Make sure flight publications (charts, chart supplement, etc.) are up to date.
- Student questions.

References

- Advisory Circulars. ([Search Tool](#))
 - Aviation Weather. ([AC 00-6B](#))
 - Aviation Weather Services. ([AC 00-45H](#))
- Aeronautical Information Manual.
 - Chapter 06 - Emergency Procedures.
- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 11 - Night Operations.
- Federal Aviation Regulations.
 - Suggested study list (Private/Recreational)
 - Focus on cross country regulations.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 12 - Weather Theory.
 - Chapter 13 - Aviation Weather Services.

Lesson Content**Night Operations**

- The student was instructed to study night operations as this was introduced during the pre-brief of lesson 133.
- Quiz the student on (refer to lesson 133 if needed):
 - Definitions of nighttime.
 - Night vision (cones and rods).
 - Night illusions.
 - Aircraft and airport lighting.
 - Preflight inspection.

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Reading Charts at Night

- Reading charts at night can be tricky due to the lack of lighting.
- A flashlight (red) must be brought on the flight.
 - Note: the red light can make red lines on the sectional hard to see.
- Chart symbology will be reviewed under the cross-country flight planning section below.

Cockpit Lighting

- The avionics, panel, and switches' lights should be kept as dim as possible to avoid night vision to be impaired.
- The student can make use of the dome light if needed. This light has a green/blue shade on it in order to avoid white light.
 - Demonstrate how to operate if needed.
- If a phone or tablet is used for any reason, ensure that it is in night mode and at minimal brightness.

Personal Equipment

- On top of all regular personal equipment, ensure that the student has the following for the night cross-country:
 - Flashlight: red and white.
 - White light is used to preflight the airplane while the red light is used while in flight.
 - Again, red light will not impair night vision.
 - Sectional chart, chart supplement, etc.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!
- At night, the use of peripheral vision is the best method to scan for traffic.
- Use the same scanning method but paying more attention to your peripheral vision. If a plane is spotted, avoid looking directly at it. Look slightly to the side of it.

Cross Country Procedures

- For the purpose of this ground, plan the 1st leg of the cross-country to be flown on lesson 135.
 - Flight planning and navigation log:
 - The student should know how to plan a flight and fill out a nav log at this point.
 - Ensure that they are drawing the airport diagrams, filling out the right frequencies, using the PIM to determine power settings, etc.
 - NOTAMs, AIM, Chart supplement:
 - Ensure the student knows how to obtain NOTAMs.
 - Quiz the student on the different types of NOTAMs.
 - Ensure the student is familiar with the various sections of the AIM.
 - Chapter 01 - air navigation.
 - Chapter 02- aeronautical lighting.
 - Chapter 03 - airspace.

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- Chapter 04 - air traffic control.
- Chapter 05 - air traffic procedures.
- Chapter 06 - emergency procedures.
- Chapter 07 - safety of flight
- Chapter 08 - medical facts.
- Chapter 09 - aeronautical charts.
- Chart supplement:
 - Ensure that the student can read and understand the desired airport's information page.
 - If they don't, see if they can reference the legend.
 - Check the types of approach lights, runway lighting, and tower operating hours!
- Route selection:
 - The student's route selection should be based on:
 - Airspace.
 - Cloud clearances.
 - Glide distance.
 - Obstacles.
 - Make sure that the student draws the course line on the sectional.
- Night checkpoints:
 - It's important to pick checkpoints that can easily be seen at night.
 - Examples:
 - Light patterns of town are easily identified.
 - Rotating beacons are useful:
 - Civil airports – White/Green.
 - Military airports – 2xWhite/Green.
 - Waterport – White/Yellow.
 - Heliport – White/Green/Yellow.
 - Highways with cars on them are usually easy to see (headlights/brake lights are visible).
- Altitude selection:
 - The student's altitude selection should be based on:
 - Airspace.
 - Cloud clearances.
 - It is difficult to see clouds at night – be cautious to avoid flying into MVFR/IFR weather conditions.
 - 1st indication – Gradual disappearance of the ground and glowing around lights.
 - Glide distance.
 - Obstacles.

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- Magnetic course.
- Alternate planning:
 - It is always a good idea to plan for an alternate in the event of a malfunction, deteriorating weather, aeromedical factors, sudden airport closure...
 - Ask the student what would be considered a good alternate airport for this flight.
- Navigation systems:
 - It's a good idea to increase the use of navigation systems at night due to the lack of visual references!
 - GPS:
 - Before taxiing, enter the route of flight into the GPS.
 - The GPS will provide information similar to the Nav Log and update distances, time enroute, fuel burn, etc. as changes are made to the flight plan.
 - This saves you, tons of work and time, especially in the case of a diversion or emergency
 - The GPS is an extremely useful tool and should be used once the pilot is capable of competently navigating with it.
 - Great for situational awareness.
 - Ensure you don't get complacent with the GPS and lose situational awareness, a GPS failure in this case could be hazardous.
 - If the GPS fails, you must be able to safely continue along the route by means of pilotage and dead reckoning.
 - VOR:
 - Use the sectional chart to figure out which radial to track.
 - Make sure the VOR frequency and radials are written on the navlog.
 - The student must tune and ID the VOR prior to using it.
 - Show on the sectional chart how to use a VOR airway/radial to get from point to point.
- Increase use of flight instruments:
 - As aforementioned, using visual references at night can be difficult.
 - Emphasize the increased use of flight instruments!
 - Attitude indicator.
 - Altimeter.
 - HSI.
 - VSI.
 - Turn coordinator.
- Diversion procedures
 - Quiz the student on diversion procedures.
 - Point at a random location on the route of flight and see how the student would go about diverting to an airport.

Regulations

- 91.103 Preflight action
 - IFR flight or away from the vicinity of an airport.
 - NOTAMs.
 - Weather.
 - Known delays.
 - Runway lengths.
 - Alternates if needed.
 - Fuel requirements.
 - Takeoff/landing data.
- 91.151 Fuel requirements (FIT and FAA)
 - Day: 30 mins.
 - Night: 45 mins.
 - FIT: 60 mins day or night.

Adverse Weather

- Fog:
 - Temperature of air near the ground is cooled to the air's dew point
 - Types (reported when visibility is less than 5/8th of a mile)
 - Radiation fog – forms in low-lying areas like mountain valleys. Occurs when the ground cools rapidly due to terrestrial radiation, and surrounding air reaches dew point. As the sun rises and temperature increases, the fog burns off
 - Advection fog – when a layer of warm air moves over a cold surface. 15 knot wind is necessary. Common to coastal areas where sea breezes can blow the air over cooler landmasses
 - Upslope fog – Occurs when moist, stable air is forced up sloping land features like a mountain range. Also requires wind for formation and continued existence
 - Steam fog – or sea smoke, forms when cold dry air moves over warm water. As water evaporates it turns into smoke
- Visibility and ceilings:
 - Stress that continued flight into deteriorating weather is inadvisable.
 - Discuss the disadvantages of scud running of VFR over the top.
 - Discuss diversion procedures.
- Wind
 - Refer to hazards posed by exceeding wind limitations.
 - Discuss diversion to a proper alternate with either calmer winds or more headwind.
 - Review the use of V_A and V_{NO} to prevent damage in turbulence.
- Turbulence
 - Types:
 - Light chop: slight rapid bumpiness.
 - Light turbulence: series of momentary changes in your altitude or attitude.

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- Moderate chop: more intense than light chop.
- Moderate turbulence: changes in attitude and altitude, positive control though.
- Severe turbulence: large abrupt changes in altitude and attitude, tempo loss of control.
- Extreme turbulence: aircraft violently tossed around, impossible to control. Cause structural damage.
- CAT → severe turbulence in cloudless regions, usually at high altitude with wind shear.
- Intensities:
 - Occasional.
 - Intermittent.
 - Continuous.
- Thunderstorms:
 - Emphasize that thunderstorms possess all known weather hazards.
 - Refer to the Flight Operations Manual for guidance (hold outside/cancel/divert).
 - What you need for a thunderstorm:
 - Lifting action.
 - High moisture.
 - Unstable air.
 - Stages:
 - Developing stage:
 - Towering cumulus (rising air), little rain, occasional lightning.
 - Mature:
 - Fully formed, heavy rain, frequent lightning, strong winds.
 - Dissipating
 - Downdrafts, rainfall decreases, still produce strong winds and lightning.
 - Types:
 - Single cell: develops on warm humid days. severe, produce hail, and microburst.
 - Multi cell: develop in clusters, cover large areas, can move in different directions.
 - Squall line: narrow band of TS, ahead of cold front (moist, unstable air, lifting action). Severe and wide.
 - Super cell: Single long-lived TS. large hail and significant tornadoes.
 - What to expect:
 - Lightning.
 - Turbulence.
 - Hail.
 - Tornadoes.
 - Wind shear.

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- Icing conditions:
 - Structural:
 - Requires visible moisture and aircraft surface below freezing.
 - Clear → forms when large supercooled drop hit the aircraft and spread and freeze slowly
 - Rime → forms when small drops hit the aircraft and freeze rapidly.
 - Mix → mix of the two
 - Effects of icing:
 - Drag → increase
 - lift → decrease
 - Weight → increase
 - Stall speed → increase
 - Induction:
 - Carb ice → can happen with temperatures as high as 100F/38C and a relative humidity of 80%.
 - Air in the carburetor is accelerated in the venturi, causing the temperature of the air to decrease. More susceptible to icing.
 - Turn carb heat on (small reduction in RPM, warm air is less dense than cold air, enriches the mixture).
 - Instrument icing:
 - Pitot tube and static ports.
 - Static wicks.
 - Antennas.
 - Stall warning tab.
 - OAT probe.

Carburetor Icing

- Give the student a scenario where the engine is running rough in a high humidity and low temperature environment.
- See if the student can correlate this scenario to carburetor icing.
- Ensure that the student can run through the correct procedure.

Lost Procedures

- 7C's:
 - Calm.
 - Climb.
 - Identify prominent landmarks.
 - Circle.
 - Avoid aggravating the situation.
 - Crosscheck.
 - GPS.

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- VOR: use 2 VOR's, center the CDI, draw on the sectional chart to determine position.
- Communicate.
 - Request assistance from ATC.
- Comply.
 - Follow ATC instructions.
- Conserve.
 - reduce power to 55% and lean the mixture for best economy (refer to PIM).

Low Fuel

- Review situations that might lead to low fuel.
 - Poor preparation, mixture too rich, deviating from POH, headwinds, etc...
- Explain that a diversion to an alternate with fuel services is best.
 - In the scenario cross country, point out suitable alternates (tick marks).

Communications or Navigation Radio Failure

- The first step is always to troubleshoot
 - Double check the volume.
 - Audio panel.
 - Comm volume.
 - Headset volume.
 - Check to make sure the headset is plugged in.
 - Ensure the frequency is set correctly.
 - Attempt using the alternate communications radio. (com1/com2)
 - Attempt to call the tower with a cell phone.
- If troubleshooting is unsuccessful, squawk 7600.
- If close to a non-towered airport, proceed there.
- If not, circle over the towered airport at 500' above traffic pattern altitude.
- Review light gun signals.

Electrical Failure

- Give the student a scenario where an electrical failure is detected to happen (decrease in alternator amps, annunciator panel light, etc.).
- Question the student on their decision making.
- Ensure that they can refer to the emergency checklist.

Instrument Failures

- Pitot/static:
 - Review the pitot/static blockages.
 - See if the student can recall pitot heat and alternate static sources.
- Vacuum:
 - Refer to the annunciator panel light and vacuum suction gauge.
 - Glass cockpit may replace vacuum instruments.

Emergency Equipment and Survival Gear

- Fire extinguisher:
 - Check in preflight to make sure it is properly charged (in the green), the pin is in, and it is latched.
 - Operation: open the latch, pull the pin out, point at the base of the fire and spray. If used in flight, make sure the vents and windows are opened.
- Survival kit:
 - A survival kit should provide sustenance, shelter, medical care and a means to summon help (radio, phone). Consider the terrain, the climate, and type of emergency communication equipment needed:
 - Winter: include blankets, extra clothes and jackets if available
 - Summer: blankets might be a good idea because it can still get cool at night
 - Mountainous terrain in December requires different survival gear than ocean flying in August.
- Overwater operations:
 - 91.509:
- Emergency locator transmitter (ELT):
 - These are designed to transmit a signal on 121.5 at a frequency of 243 megahertz (406 on newer ELTs).
 - 121.5: sounds like a loud beeping noise. The closer you are, the louder the sound.
 - 243/406 mhz: sends a signal containing the aircraft's location, pilot's name, contact information, and type of aircraft.
 - Usually armed to go off at an impact of around 5G's. Can also be manually turned on in the cockpit (ARM vs ON).
 - Batteries must be replaced after one hour of cumulative use or half the useful life is up.
 - Inspection is every 12 calendar months.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Schedule the next lesson.
- Preview the next lesson → 135.
- Assign homework:
 - Plan the cross-country to be flown in lesson 135.

Completion Standards

This lesson is complete when the student:

1. Has been introduced to all elements of night cross-country planning and procedures and can plan a night cross-country flight with instructor guidance.